P6: Using Learning to Create a Novel Game Controller

Section 8: Transfer Learning

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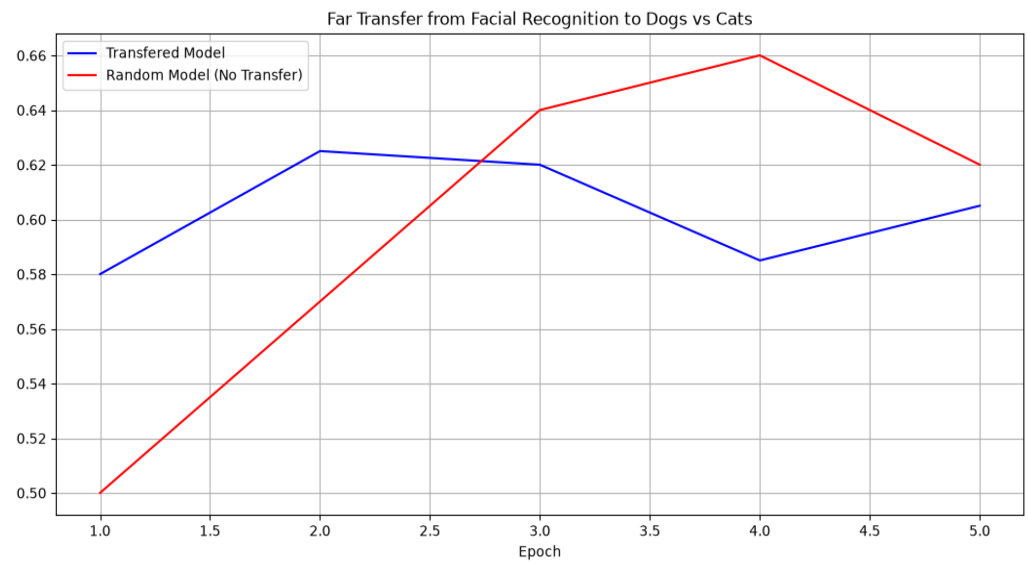
Far Transfer from Facial Recognition to Dogs vs Cats Classification

Validation Accuracy: With Transfer vs. Without Transfer



The transfer model starts at a substantially higher validation accuracy (58.0% at epoch 1) since it only needs to train a small classification head on top of already useful frozen features from the facial recognition network. The randomly initialized model, by contrast, starts at exactly chance level (50.0%) since it must learn its entire architecture from scratch. This increased intercept after epoch 1 is one of the accepted criteria for demonstrating a positive transfer effect. Both models reach similar accuracy by epoch 3, after which the random model's larger number of trainable parameters allows it to keep improving, while the transfer model's small head of only 4,290 trainable parameters plateaus. This tradeoff is expected: the frozen features give transfer an immediate head start, but limit how far it can climb given only 5 epochs of training on a small dataset.

Final Test Set Accuracy

Model	Test Accuracy
Transfered Model (with transfer)	54.67%
Random Model (without transfer)	55.00%

Final test accuracy is close between the two models (54.67% vs 55.00%). As with the validation curve, the transfer effect shows up primarily as a faster and higher starting early trajectory rather than a final endpoint advantage after 5 epochs.

Facial Recognition Classes

Rows top to bottom: neutral, happy, surprise

Transfer Task Classes (Dogs vs. Cats)

Rows top to bottom: dogs, cats

Model Summary: Transfered Model

Frozen facial recognition base (non trainable) plus new trainable head

Layer (type)	Output Shape	Param #
sequential (frozen base)	(None, 64)	142,464
dense (Dense)	(None, 64)	4,160
dropout (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 2)	130

Total params: 146,754 (573.26 KB)   Trainable params: 4,290 (16.76 KB)   Non trainable params: 142,464 (556.50 KB)

Model Summary: Random Model (No Transfer)

Same architecture, randomly initialized and fully trainable

Layer (type)	Output Shape	Param #
sequential_2 (base)	(None, 64)	142,464
dense_2 (Dense)	(None, 64)	4,160
dropout_1 (Dropout)	(None, 64)	0
dense_3 (Dense)	(None, 2)	130

Total params: 146,754 (573.26 KB)   Trainable params: 146,754 (573.26 KB)   Non trainable params: 0 (0.00 B)